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A reconstruction method for graph signals based on the power spectral density estimation

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ABSTRACT

Graph signal reconstruction is a classic problem of graph signal processing. The ultimate goal of signal reconstruction is to obtain an estimate as close as possible to the original signals. Most of existing reconstruction models are based on assumption of signal smoothness. This paper proposes a new reconstruction model based on signal power spectral density estimation. The basic idea is to suppress those graph frequency components with low power spectral density when reconstructing a signal, while preserving those graph frequency contents with high power spectral density. Therefore, the reconstructed signals can well maintain the frequency structures of the original signals. Under the stationarity assumption of stochastic graph signals, we further propose a method to estimate the power spectral density directly from the sampled signals. Experimental results show that the proposed power spectral density estimation algorithm works well even when the proportion of labeled vertices is extremely low, and is robust to the PSD of signals itself and the various structures of graphs. Compared with traditional models based on smoothness assumption, the proposed reconstruction model can significantly improve the reconstruction performance.

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1. Introduction

Graph signal processing is an emerging research field [1–3]. It expands the theories and methods of classical signal processing, and has been well applied in practical problems. In recent years, graph signal processing and related issues have attracted the attention of many scholars. Many issues have been studied, including graph filtering [4–10], graph wavelet [11–16], graph Fourier transforms [17–20], windowed graph Fourier transform [21,22], sampling and reconstruction [23–32], as well as stationary graph signal processing [33,34], etc.

Signal reconstruction is a classical problem in signal processing. Under certain conditions, through reconstruction, all the information of a continuous signal can be recovered from its discrete sampled signal. Similarly, graph signal reconstruction also aims to reconstruct the original signal through partial sampling. There has been some theoretical analysis on the sampling and reconstruction of bandlimited graph signals [17,35–41].

In classical situations, signal reconstruction is often based on some assumptions. Similarly, the reconstruction of image signals also needs to be based on certain assumptions, among which the most commonly used assumptions are smoothness and/or band-limited assumptions. Some reconstruction methods based on the assumptions have been proposed [25,39,42]. It should be said that these assumptions are reasonable and can meet the requirements of most applications.

We know that the ultimate goal of signal reconstruction is to obtain an estimate as close as possible to the original signal. Therefore, the reconstructed signal should follow the frequency composition of the original signal. This means that when reconstructing a signal, those frequency components with lower power spectral density (PSD) in the original signal should be suppressed and those with higher PSD should be preserved.

Literature [34] proposes a reconstruction method based on the assumption of stationarity of graph signal, which is called the graph Wiener optimization framework. The authors use a so-called Fourier penalty weight to punish the frequency bands with lower signal-to-noise ratio (SNR), and propose a method to estimate the PSD of a graph signal. However, their method cannot directly estimate the PSD from the sampled signals, but must be estimated by some other realizations of the stationary stochastic graph sig-

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nal. Therefore, in the absence of other realizations, their method has limitations.

Inspired by the idea of [34], we propose a graph signal reconstruction method based on PSD estimation. The new model penalizes those frequency bands with lower PSD in the original signal so that the reconstructed signal keeps the spectral structure of the original signal as much as possible. Under the stationarity assumption of graph signals, we further propose a method to estimate the PSD directly from sampled signals. The main contributions include:

- A new algorithm to estimate the PSD of a stationary stochastic graph signal directly from sampled signals is designed;
- A reconstruction model of graph signals based on PSD estimation is proposed.

We conduct experiments on several simulation data sets and two real data sets to verify the effectiveness of the proposed methods. Experimental results show that the proposed algorithm can estimate PSD well even when the ratio of labeled vertices (Please see Section 2.1 for the definition) in sampled signals is low, and is suitable for diverse signals, different connection structure of graph and size of graph. The proposed reconstruction model has significantly better reconstruction performance than the traditional methods based on smoothness assumptions.

The rest of the paper is organized as follows: Section 2 contains the most important content of this paper. In this section, we discuss the general reconstruction model, establish a new reconstruction model based on PSD estimation, propose a new algorithm to estimate the PSD of a stationary stochastic graph signal is presented and also discuss the computational complexity of the proposed reconstruction model. In Section 3, we first verify the effectiveness of the proposed PSD estimation method through simulation experiments, and then compare the reconstruction performance of the proposed model and the general model on the simulation data sets. Finally, the reconstruction experiments are carried out on two real data sets and the experimental results are compared with some existing methods. Finally, the conclusions are given in Section 4.

2. The reconstruction model based on the PSD of signals on graph

2.1. The general model

Consider an undirected, weighted graph $\mathcal{G} := (\mathcal{V}, \mathbf{W})$ with the vertex set $\mathcal{V} = \{\mathbf{v}_i | \mathbf{v}_i \in \mathcal{R}^m, i = 0, 1, \cdots, N-1\}$ and the weighted adjacency matrix $\mathbf{W}$ whose entry w_{ij} reflects the similarity between the vertices $\mathbf{v}_i$ and $\mathbf{v}_j$, where $\mathbf{v}_i$ is the feature vector representing the vertex. The graph Laplacian is defined as $\mathbf{L} := \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the diagonal degree matrix with the diagonal element $d_i = \sum_{j=0, j\neq i}^{N-1} w_{i,j}$ for $i = 0, 1, \cdots, N-1$. In many problems, we also assign a value y_i to each vertex, which is called the vertex's label. Sort the vertices, and call the vector $\mathbf{y} = (y_0, y_1, \cdots, y_{N-1})^T$ formed by their corresponding labels a signal on the graph or graph signal. The reconstruction problem of graph signal is: Assuming we have obtained the labels on some vertices, how to estimate the labels on other vertices? For the sake of convenience, in the following, we call this kind of signal the sampled signal that only the labels on part of vertices are known. Vertices with known labels are called labeled vertices, and vertices with unknown labels are called unlabeled vertices.

If we can find a function defined on the vertex set $\mathcal{V}$ whose function values on the labeled vertices are equal to the known labels, then its function values on the unlabeled vertices can be served as estimates of the labels of the unlabeled vertices. Therefore, the reconstruction problem of graph signals can be modeled as to find a function f such that

$$f(\mathbf{v}_i) = y_i \quad \forall \mathbf{v}_i \in \mathcal{K}, \tag{1}$$

where $\mathcal{K} \subset \mathcal{V}$ is the set of labeled vertices, namely $\mathcal{K} = \{\mathbf{v}_i |$ if y_i is known$\}$.

In general problems, we always assume that graph signals have a certain degree of smoothness. The measure usually used to describe the smoothness of graph signals is

$$\rho(f) := \frac{1}{2} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} w_{i,j} (f(\mathbf{v}_i) - f(\mathbf{v}_j))^2. \tag{2}$$

Since $w_{i,j}$ describes the similarity relationship between the vertices $\mathbf{v}_i$ and $\mathbf{v}_j$, the physical meaning of this smoothness measure is that if two vertices are strongly connected, their signal values should be relatively close.

After introducing the smoothness constraint, the reconstruction problem of graph signals can be formulated as the following constrained optimization problem:

$$\min_{f:\mathcal{V}\to\mathcal{R}} \rho(f), \quad \text{subject to} \quad f(\mathbf{v}_i) = y_i, \ \ \mathbf{v}_i \in \mathcal{K}. \tag{3}$$

It can be approximated to the following unconstrained optimization objection by introducing a penalty term

$$\min_f r(f) := \sum_{\mathbf{v}_i \in \mathcal{K}} (f(\mathbf{v}_i) - y_i)^2 + \beta \rho(f). \tag{4}$$

The first term on the right hand side of the above formula is called the approximation term, and it makes the value of solution f at the vertex $\mathbf{v}_i$ is approximately equal to the known label y_i. The second term is called the penalty term, and its function is to make the solution f as smooth as possible on the graph. The parameter $\beta > 0$ is called the regularization parameter, which is used to adjust the weight of approximation term and penalty term in the objective function $r(f)$. Depending on its value, the solution has different smoothness. It should be noted that the solution of Equation (4) usually does not satisfy the constraints of Equation (3), so usually it is not a solution of Equation (3), but just an approximation of it.

Extract the rows corresponding to the labeled vertices from the identity matrix $\mathbf{I}_{N\times N}$ to form a matrix with size of $k \times N$, denoted as $\mathbf{E}_k$, where $k := |\mathcal{K}|$ is the number of labeled vertices, then the approximation term in Equation (4) can be written as:

$$\sum_{\mathbf{v}_i \in \mathcal{K}} (f(\mathbf{v}_i) - y_i)^2 = \|\mathbf{E}_k \mathbf{f} - \mathbf{y}_k\|^2, \tag{5}$$

where $\mathbf{f} = (f(\mathbf{v}_0), f(\mathbf{v}_1), \cdots, f(\mathbf{v}_{N-1}))^T$, $\mathbf{y}_k$ is the vector consisting of the known labels. The right side of Equation (2) can be calculated as follows

$$\begin{aligned}
&\frac{1}{2} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} w_{i,j} (f(\mathbf{v}_i) - f(\mathbf{v}_j))^2 \\
&= \frac{1}{2} \sum_{i=0}^{N-1} \left(\sum_{j=0}^{N-1} w_{i,j} \right) f^2(\mathbf{v}_i) + \frac{1}{2} \sum_{j=0}^{N-1} \left(\sum_{i=0}^{N-1} w_{i,j} \right) f^2(\mathbf{v}_j) \\
&\quad - \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} w_{i,j} f(\mathbf{v}_i) f(\mathbf{v}_j) \\
&= \frac{1}{2} \sum_{i=0}^{N-1} d_i f^2(\mathbf{v}_i) + \frac{1}{2} \sum_{j=0}^{N-1} d_j f^2(\mathbf{v}_j) - \mathbf{f}^T \mathbf{W} \mathbf{f}
\end{aligned}$$

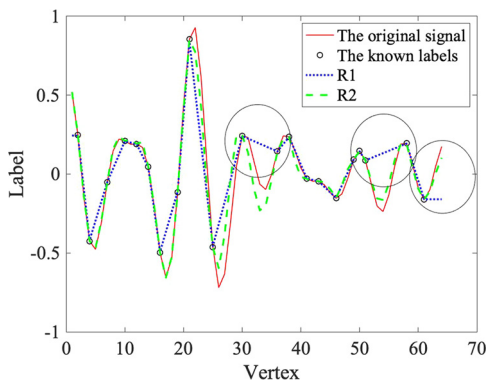

Fig. 1. An example of reconstruction of signals on graph. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$
\begin{aligned}
&= \frac{1}{2}\mathbf{f}^T\mathbf{D}\mathbf{f} + \frac{1}{2}\mathbf{f}^T\mathbf{D}\mathbf{f} - \mathbf{f}^T\mathbf{W}\mathbf{f} \\
&= \mathbf{f}^T(\mathbf{D}-\mathbf{W})\mathbf{f} \\
&= \mathbf{f}^T\mathbf{L}\mathbf{f},
\end{aligned} \tag{6}
$$

where $d_i = \sum_{j=0, j\neq i}^{N-1} w_{i,j}$ is the degree of the vertex $\mathbf{v}_i$ and $\mathbf{D}$ is the diagonal degree matrix with the diagonal element d_i for $i = 0, 1, \cdots, N-1$. By combining Equation (5) and Equation (6), the optimization problem (4) can be written as:

$$\min_{\mathbf{f}} r(\mathbf{f}) := \|\mathbf{E}_k\mathbf{f} - \mathbf{y}_k\|^2 + \beta\mathbf{f}^T\mathbf{L}\mathbf{f}. \tag{7}$$

For convenience, in the following we call the reconstruction model of Equation (7) the general model, referred to as GM for short. It can be solved analytically by solving equation

$$(\mathbf{E}_k^T\mathbf{E}_k + \beta\mathbf{L})\mathbf{f} = \mathbf{E}_k^T\mathbf{y}_k. \tag{8}$$

Once $\mathbf{f}$ is obtained, $f(\mathbf{v}_i)$ are the estimates of the labels on the unlabeled vertices $\mathbf{v}_i \notin \mathcal{K}$.

2.2. The reconstruction model based on PSD of graph signals

The penalty term in the above model punishes high frequency contents and protects low frequency components, and lacks adaptability. Therefore, it cannot guarantee that the reconstructed signals follow the frequency structures of the original signals. Let us look at the example in Fig. 1. The red solid curve is a signal defined on a Path graph with 64 vertices. The markers '∘' represent the labeled vertices which are used to reconstruct the signal. The blue dotted curve indicates the reconstructed signal by using the above penalty term. Since it does not have to follow the frequency structure of the original signal, but only needs to make the smoothness measure as small as possible, therefore, the waveform of the reconstructed signal cannot fit the waveform of the original signal well. Some areas that are poorly fitted have been marked with circles in the figure to make them easier for readers to identify. It is worth noting that the smoothness mentioned here does not mean that the function is derivable as that in classical situations, but that the smoothness measurement of Equation (2) is as small as possible. The green dashed curve is another reconstruction. Intuitively, one has no difficulty seeing that it is a better estimation to the original signal, that is, its waveform fit to that of the original signal well.

Now we begin a detailed discussion of how to use the frequency structure of the original signals to reconstruct graph signals. Since the Laplacian $\mathbf{L}$ is a symmetric positive semi-definite matrix, it has a complete set of orthogonal eigenvectors, denote them as $\mathbf{U} = (\mathbf{u}_0, \ldots, \mathbf{u}_{N-1})$, and called the Fourier basis. Their corresponding eigenvalues are a set of non-negative real numbers. For convenience, we sort them as follows: $0 = \lambda_0 < \lambda_1 \leqslant \cdots \leqslant \lambda_{N-1}$, then $\mathbf{L}$ can be written as:

$$\mathbf{L} = \mathbf{U}\Lambda\mathbf{U}^T, \tag{9}$$

where Λ is a diagonal matrix with $\lambda_\ell (\ell = 0, \cdots, N-1)$ as the diagonal elements. The Fourier transform of the graph signal $\mathbf{f}$ is defined as

$$\hat{\mathbf{f}} := \mathbf{U}^T\mathbf{f}. \tag{10}$$

The inverse Fourier transform is

$$\mathbf{f} := \mathbf{U}\hat{\mathbf{f}}. \tag{11}$$

Let us look at the penalty term in Equation (6) again, which can be written as

$$\rho(f) = \mathbf{f}^T\mathbf{L}\mathbf{f} = \mathbf{f}^T\mathbf{U}\Lambda\mathbf{U}^T\mathbf{f} = \hat{\mathbf{f}}^T\Lambda\hat{\mathbf{f}} = \sum_{\ell=0}^{N-1}\lambda_\ell\hat{\mathbf{f}}^2(\ell). \tag{12}$$

We keep the name in classical signal processing and call $\hat{\mathbf{f}}^2(\ell)$ the energy which the graph signal $\mathbf{f}$ possesses the frequency component λ_ℓ. The penalty term in Equation (6) is actually a weighted sum of the energies of various frequency components in signals. The weighting factors are the graph frequencies $\lambda_\ell (\ell = 0, \cdots, N-1)$. This means that the higher the frequency, the greater the penalty imposed, and vice versa. Since the energy of most signals is mainly concentrated in the lower frequency bands, this strategy is reasonable when we do not have any priori knowledge of the original signal. However, if the PSD of a signal is known, then the penalty term cannot well represent the reconstruction target, which motivates scholars to consider variations of the penalty term [43,44].

We know that the ultimate goal of reconstructing a signal is to obtain an estimate that is as close as possible to the original signal. Therefore, the penalty term should try to preserve the frequency contents with higher PSD and impose large penalties on those with lower PSD. Based on this idea, we modify the penalty item as follows:

$$\rho(f) = \sum_{\ell=0}^{N-1} w(\gamma_{\mathbf{f}}(\lambda_\ell))\hat{\mathbf{f}}(\ell)^2 = \hat{\mathbf{f}}^T\Lambda_w\hat{\mathbf{f}}, \tag{13}$$

where $\gamma_{\mathbf{f}}(\lambda_\ell)$ is the PSD corresponding to the frequency λ_ℓ, $w(x)$ is a monotonic decreasing function of x, for convenience, in the following, we call it the weight function of PSD, referred to as weight function, and Λ_w is a diagonal matrix with $w(\gamma_{\mathbf{f}}(\lambda_\ell))(\ell = 0, \cdots, N-1)$ as the diagonal elements.

Looking back at the penalty term in Equation (12), it always takes the graph frequency as the weighting factor. This means that the weighting factor has nothing to do with the frequency content of the original signal itself. The modified penalty term is related to the PSD of the original signals. For different signals, it takes different weighting factor. Therefore, we can regard it as an adaptive penalty item. Finally, we get a new reconstruction model as follows:

$$\min_{\mathbf{f}} r(\mathbf{f}) := \|\mathbf{E}_k\mathbf{f} - \mathbf{y}_k\|^2 + \beta\mathbf{f}^T\mathbf{U}\Lambda_w\mathbf{U}^T\mathbf{f}. \tag{14}$$

It also can be solved analytically by solving the equation:

$$(\mathbf{E}_k^T\mathbf{E}_k + \beta\mathbf{U}\Lambda_w\mathbf{U}^T)\mathbf{f} = \mathbf{E}_k^T\mathbf{y}_k. \tag{15}$$

2.3. PSD estimation based on sampled signals

In the above discussion, we propose a modified penalty term based on the assumption that the PSD is known. However, in practical problems, the PSD of sampled signals is often unknown. Therefore, the next problem we face is how to estimate the PSD through sampled signals.

In classical signal processing, the PSD estimation is a basic problem, and many estimation methods have been reported. The typical representatives are the methods based on the windowed Fourier transform which are proposed by Bartlett [45] and Welch [46]. Barlett's method can be summarized as follows: first remove the mean, then cut the signal into equal and non-overlapping segments, then calculate the Fourier transform of each segment, and finally find the average of the square of the Fourier coefficient's amplitude of all segments, and the result is considered as the estimate of the PSD. Welch's method is a generalization of Barlett's method, which allows overlap between segments. Inspired by the ideas of Bartlett and Welch, Perraudin [34] et al. proposed a PSD estimation method of graph signals. Their method is based on the windowed Fourier transform on graph [47], but instead of a translated rectangular window in time (space), they use a kernel g shifted by multiples of a step τ in the spectral domain, i.e.

$$g_m(\lambda_\ell) = g(\lambda_\ell - m\tau), \quad m = 1, \cdots, M, \quad \tau = \frac{\lambda_{max}}{M},$$

where M is the number of given frequency bands. They then localize the 'window' at each individual vertex of graphs. A filter window for the mth frequency band locating at the vertex $\mathbf{v}_i$ is defined as $\mathcal{T}_i g_m$, where $\mathcal{T}_i$ is the localization operator which translates the filter window to the vertex $\mathbf{v}_i$. The coefficients of the graph windowed Fourier transform of the graph signal x can be written as a matrix with elements

$$C[i, m] = < x, \mathcal{T}_i g_m > . \tag{16}$$

The average of the squared coefficients of this transform over the vertex set is considered as the estimate of the PSD.

Under the stationarity assumption of graph signals, the method of Perraudin et al. can estimate the PSD of a graph signal well. However, their method is proposed under such an assumption that we have a few of realizations of the stationary stochastic graph signal in hand. When the graph signal x is a sampled signal, since the labels of the unlabeled vertices are set to 0, only the labels of the labeled vertices have contribution to the coefficients of the graph windowed Fourier transform. Therefore a large error will be inevitable in the estimated PSD.

It is obviously impossible to accurately estimate the PSD of a stationary stochastic graph signal only by a single sampled signal, especially when the proportion of labeled vertices is very low. However, it may be feasible, if the number of sampled signals is large enough, and they satisfy the assumptions as follows:

1. All sampled signals come from realizations of a stationary stochastic graph signal;
2. Each sampling is independent and stochastic.

Fortunately, in many practical applications, these assumptions can be approximately satisfied. For example, in the recommendation system, the user's ratings of items can be considered as sampled signals that meet the above assumptions.

Below, we discuss how to estimate the PSD of a stationary stochastic graph signal directly from sampled signals. Let $\mathbf{f}$ be a stationary stochastic signal defined on the graph $\mathcal{G}$, according to the properties of the stationary graph signal, we have

$$\mathbb{E}\{f_i\} = c \in \mathcal{R}, \tag{17}$$

where f_i is a random variable representing the label value on the vertex $\mathbf{v}_i$. The PSD of $\mathbf{f}$ can be written as [34]:

$$\gamma_{\mathbf{f}}(\lambda_\ell) = \mathbf{u}_\ell^T \Sigma_{\mathbf{f}} \mathbf{u}_\ell \ \ (\ell = 0, \cdots, N-1), \tag{18}$$

where $\Sigma_{\mathbf{f}}$ is the covariance matrix of $\mathbf{f}$. It can be expanded as

$$\gamma_{\mathbf{f}}(\lambda_\ell) = \sum_{j=0}^{N-1} u_{\ell j}^2 \Sigma_{\mathbf{f}}(j, j) + 2 \sum_{i,j=0, i\neq j}^{N-1} \Sigma_{\mathbf{f}}(i, j) u_{\ell i} u_{\ell j}, \tag{19}$$

where $\Sigma_{\mathbf{f}}(j, j)$ is the variance of f_j, and $\Sigma_{\mathbf{f}}(i, j)$ is the covariance of f_i and f_j. Equation (19) tells us that if we can estimate $\Sigma_{\mathbf{f}}(j, j)$ and $\Sigma_{\mathbf{f}}(i, j)$ by some sampled signals, then we can approximately calculate the PSD of $\mathbf{f}$.

Let $\mathbf{M} = (m_0, \cdots, m_{N-1})^T$ be the vector which consists of N random variables m_i $(i = 0, \cdots, N-1)$ which are independent and obey the Bernoulli distribution with parameter p, namely

$$P(m_j = 1) = p, \ P(m_j = 0) = 1 - p \ \ (j = 0, \cdots, N-1). \tag{20}$$

In addition, we consider that the sampled signals are affected by random white noise ϵ which obeys a Gaussian distribution with a mean of 0 and a variance of σ^2. Then, a sampled signal can be expressed as $\mathbf{M} \circ (\mathbf{f} + \epsilon)$, where the symbol $'\circ'$ represents the Hadamard product of matrix. We have

$$\mathbb{E}(m_j(f_j + \epsilon_j) = pc \quad (j = 0, \cdots, N-1) \tag{21}$$

$$\begin{aligned}
\Sigma_{M\circ(\mathbf{f}+\epsilon)}(j, j) &= \mathbb{E}\{(m_j(f_j + \epsilon_j))^2\} - \mathbb{E}^2\{m_j(f_j + \epsilon_j)\} \\
&= \mathbb{E}\{(m_j^2(f_j^2 + \epsilon_j^2)\} - (pc)^2 \\
&= P(m_j = 1)\mathbb{E}\{(f_i^2 + \epsilon_j^2)\} - (pc)^2 \\
&= p(\Sigma_{\mathbf{f}}(j, j) + c^2 + \sigma^2) - (pc)^2 \\
&= p\Sigma_{\mathbf{f}}(j, j) + p(1-p)c^2 + p\sigma^2.
\end{aligned} \tag{22}$$

$$\begin{aligned}
\Sigma_{M\circ(\mathbf{f}+\epsilon)}(i, j) &= \mathbb{E}\{(m_i(f_i + \epsilon_i) - pc)(m_j(f_j + \epsilon_j) - pc)\} \\
&= \mathbb{E}\{m_i m_j (f_i + \epsilon_i)(f_j + \epsilon_j)\} - (pc)^2 \\
&= P(m_i = 1)P(m_j = 1)\mathbb{E}\{f_i f_j + f_i\epsilon_j + f_j\epsilon_i \\
&\quad + \epsilon_i\epsilon_j\} - (pc)^2 \\
&= p^2\mathbb{E}\{f_i f_j\} - (pc)^2 \\
&= p^2(\Sigma_{\mathbf{f}}(i, j) + c^2) - (pc)^2 \\
&= p^2\Sigma_{\mathbf{f}}(i, j) \quad (i \neq j).
\end{aligned} \tag{23}$$

Combining (19), (22) and (23), we have

$$\begin{aligned}
\gamma_{\mathbf{f}}(\lambda_\ell) = &\frac{1}{p}\sum_{j=0}^{N-1} u_{\ell j}^2 \left[\Sigma_{M\circ(\mathbf{f}+\epsilon)}(j, j) - p(1-p)c^2 - p\sigma^2\right] \\
&+ \frac{2}{p^2}\sum_{i,j=0, i\neq j}^{N-1} \Sigma_{M\circ(\mathbf{f}+\epsilon)}(i, j) u_{\ell i} u_{\ell j}.
\end{aligned} \tag{24}$$

So far, we have known how to estimate the PSD of a stationary stochastic graph signal directly from the sampled signals. Finally, we summarize the estimation algorithm of PSD as follows:

Algorithm 1 *(PSD estimation). Let $\mathcal{G}$ be a graph with N vertices, $\mathbf{U}$ is its Fourier basis, $\mathbf{Y} = (\mathbf{y}_1, \cdots, \mathbf{y}_M)$ is a matrix each column of which represents a sampled signal on graph, $\mathcal{Y}$ is the set of all known labels of $\mathbf{Y}$, σ^2 is the given variance of random white noise.*

step 1 *Set the labels of all unlabeled vertices to 0;*

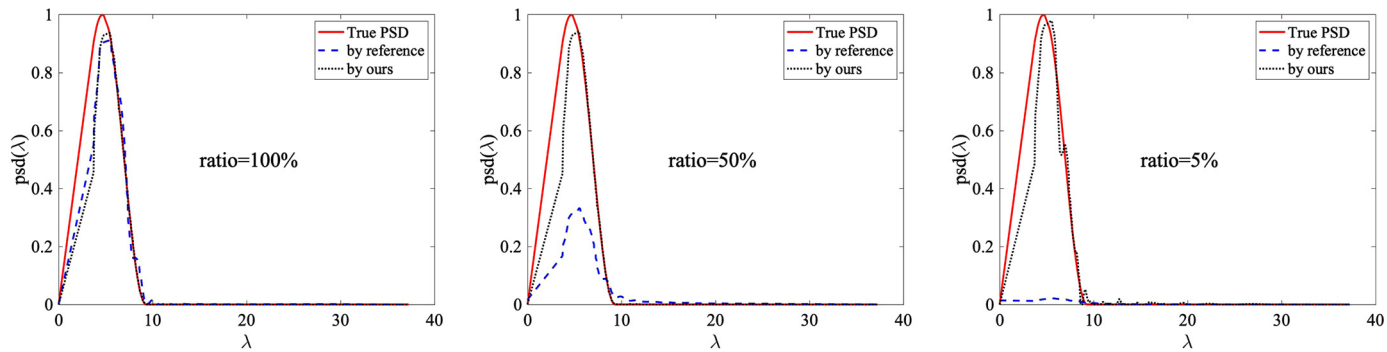

Fig. 2. PSD estimation using sampled signals with different ratios of labeled vertices.

step 2 *Calculate* $c = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}\mathbf{y}_i(j)}{|\mathcal{Y}|}$ *and* $p = \frac{|\mathcal{Y}|}{NM}$, *where* $\mathbf{y}_i(j)$ *is the label on the* j*th vertex of the* i*th sampled signal,* $|\mathcal{Y}|$ *means the number of elements in* $\mathcal{Y}$*;*

step 3 *Calculate the covariance matrix of* $\mathbf{Y}$*;*

step 4 *Calculate* $\gamma_{\mathbf{f}}(\lambda_\ell)$ $(\ell = 0, \cdots, N-1)$ *according to Equation* (24).

2.4. Computational complexity

Let us discuss the computational complexity of the proposed reconstruction method. The whole calculation can be divided into two parts: Solving the reconstruction model, and estimating the PSD. For the sake of simplicity, we only consider multiplication and division operations.

According to Equation (15), the main calculations of solving the reconstruction model include that 1) Compute the eigen decomposition of graph Laplacian $\mathbf{L}$ to get $\mathbf{U}$, the computational complexity is $O(N^3)$; 2) Compute $\mathbf{E}_k^T\mathbf{E}_k$, the amount of calculation is N^2k, where k is the number of labeled vertices; 3) Calculate $\beta\mathbf{U}\Lambda_w\mathbf{U}^T$, the calculation amount is N^3+2N^2; 4) Calculate $\mathbf{E}_k^T\mathbf{y}_k$, the calculation amount is Nk; 5) Using the well-known Gaussian elimination method to solve Equation (15), the computational complexity is $O(N^3/3)$. To sum up, the computational complexity for solving the reconstruction model is about $O(7N^3/3)$.

To estimate the PSD, we need: 1) Compute covariance matrix, the amount of calculation is $(M+1)(N^2+N)/2$, where M is the number of sampled signals; 2) Calculate the PSD according to Equation (24). The calculation amount is $N+1+2(N^2-N)+1 = 2N^2-N+2$. Therefore the computational complexity for estimating the PSD is $O((M+5)N^2/2+(M-1)N/2)$.

Compared with GM, first of all, GM does not need to estimate PSD, so its calculation amount does not include the part of PSD estimation. In addition, GM does not need to calculate the eigen decomposition of graph Laplacian $\mathbf{L}$ and $\beta\mathbf{U}\Lambda_w\mathbf{U}^T$, so its computational complexity amount is about $O(N^3/3)$.

The calculation amount discussed above is that of a single sampled signal. However, when considering the reconstruction of M sampled signals, the required calculation amount is not M times the calculation amount of a single signal reconstruction. The reason is that estimating PSD, calculating the eigen decomposition of $\mathbf{L}$ and calculating $\beta\mathbf{U}\Lambda_w\mathbf{U}^T$ all only need to be performed once. Therefore, when considering the reconstruction of the M sampled signals, the comparison on calculation amount between GM and the proposed method is $O(MN^3/3)$ vs $O(MN^3/3)+O(2N^3+(M+5)N^2/2)$, not $O(MN^3/3)$ vs $O(7MN^3/3+M(M+5)N^2/2+M(M-1)N/2)$.

3. Experimental results and analysis

The experiments are divided into two parts. The first part is used to verify the effectiveness of the PSD estimation algorithm, and the second part is used to test the performance of the proposed reconstruction model. Part of the codes used in the experiments comes from the literatures [34] and [48].

3.1. Experiments of PSD estimation

Since variance estimation is not the focus of this manuscript, in the following experiments, we always assume that the signals are noise-free, that is, assuming $\sigma = 0$ in Algorithm 1.

Experiment 1: Create a Nearest neighbor graph with the number of vertices 500 and the number of the nearest neighbors 40. Use the following PSD function

$$\gamma_{\mathbf{f}}(x) = \begin{cases} sin(4\pi x/\lambda_{max}) & x < \lambda_{max}/4, \\ 0 & x \geqslant \lambda_{max}/4 \end{cases} \tag{25}$$

to randomly generate 1000 signals. They are treated as realizations of a stationary stochastic graph signal with the PSD $\gamma_{\mathbf{f}}(x)$. λ_{max} in Equation (25) is the maximum eigenvalue of the Laplacian matrix of the graph. Randomly extract labeled vertices according to the proportions of 100%, 50% and 5%, other vertices are regarded as unlabeled vertices. Set the labels of unlabeled vertices to 0. Use Algorithm 1 to estimate the PSD, and compare it with the true one and the result estimated by the reference method in [34]. The experimental parameters of the reference method are set as follows: The number of filters is 50, and the kernel function is $\hat{g}_\ell(x) = e^{-(x-\lambda_\ell)^2/\sigma^2}$ $(\ell = 0, \cdots, N-1)$. Experimental results are shown in Fig. 2, in which the red solid line and the blue dashed line are the true PSD and that estimated by the reference method respectively, the black dotted line indicates the estimated result by our method. From left to right, the proportions of labeled vertices are 100%, 50% and 5%, respectively.

It is not difficult to see from the experimental results:

1. When all vertices are labeled vertices, all the two methods receive great estimation results;
2. As the proportion of labeled vertices decreases, the estimation result of the reference method is getting worse and worse, while that of the proposed method is almost unaffected. Even if the proportion of label vertices is only 5%, our estimation result still close to the true PSD well.

Experiment 2: This experiment will test the robustness of the proposed algorithm to the diversity of signals and the different structures of graphs. The experiments are carried out on three commonly used graphs, namely Path graph, Sensor graph, and Nearest neighbor graph (the number of neighbors is 40). The number of vertices is still 500. The two PSD functions of Equations (26) and (27) are used to generate 1000 signals on each graph. The proportion of labeled vertices is fixed at 5%, other parameters are the same as the previous experiment.

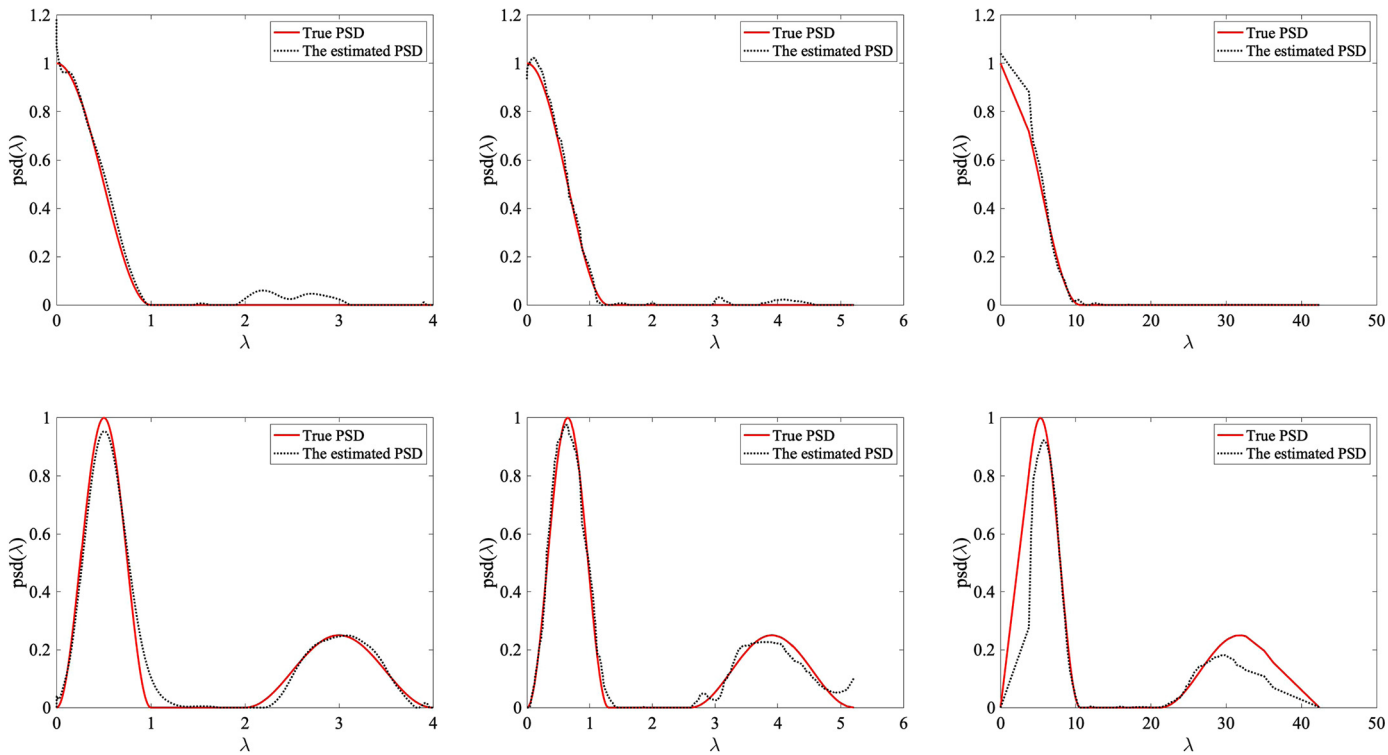

Fig. 3. PSD estimation on Path graph, Sensor graph and Nearest neighbor graph when the true PSD are $psd_1(x)$ and $psd_2(x)$ respectively. Top: $\gamma_{\mathbf{f}}^1(x)$. Bottom: $\gamma_{\mathbf{f}}^2(x)$. Left: Path graph. Middle: Sensor graph. Right: Nearest neighbor graph.

$$\gamma_{\mathbf{f}}^1(x) = \begin{cases} cos(2\pi x/\lambda_{max}) & x < \lambda_{max}/4, \\ 0 & x > \lambda_{max}/4, \end{cases} \tag{26}$$

$$\gamma_{\mathbf{f}}^2(x) = \begin{cases} sin(4\pi x/\lambda_{max}) & x < \lambda_{max}/4, \\ 0 & \lambda_{max}/4 < x < \lambda_{max}/2, \\ 0.5 * sin(2\pi x/\lambda_{max}) & x \geqslant \lambda_{max}/2, \end{cases} \tag{27}$$

Experimental results are shown in Fig. 3, the red solid line is the true PSD, and the black dotted line is the estimated result. Regardless of the PSD of signals itself and types of graphs, the estimated PSD can always close to the true one, has excellent robustness to both signals and structure of graphs.

Experiment 3: This experiment is designed to verify the effectiveness of the proposed algorithm on graphs with different sizes. We create three Nearest neighbor graphs with 1000, 2000 and 5000 vertices respectively, and use the PSD function of Equation (25) to randomly generate 1000 signals on each graph. Randomly extract 5% vertices as the labeled vertices, and others are considered as the unlabeled ones and set their labels to 0. The estimation results are shown in Fig. 4. It can be seen that our method performs well on all the graphs with different sizes.

Experiment 4: Below we will examine the influence of the number of sampled signals on PSD estimation. Create a Nearest neighbor graph with 500 vertices, and use the PSD function of Equation (25) to generate 5, 10, 20, 50, 100, 1000 signals. Randomly extract 5% vertices as the labeled vertices, and the rest of vertices are considered as the unlabeled ones and set their labels to 0. Experimental results are shown in Fig. 5. We see that when there are only 5 and 10 sampled signals, there are large errors between the estimated PSDs and the true functions. However, as the number of sampled signals increases, the estimation result is getting better and better. After the number of sampled signals is greater than 20, the estimated PSD can close to the true function well.

In the above experiments, we test the proposed estimation algorithm from multiple aspects. The following conclusions can be summarized:

1. When there are enough sampled signals, the proposed algorithm has excellent adaptability to sizes of graph, connection types of graph and frequency spectral composition of signals;
2. Even if the proportion of labeled vertices in sampled signals is very low, PSD can be estimated well.

3.2. Signal reconstruction experiments

In the following reconstruction experiments, we choose the weight function as:

$$w(x) = e^{-\alpha x}, \tag{28}$$

where α is a constant. The popular measure of accuracy — the mean absolute error (MAE) — is considered as the performance metric. It is defined as follows:

$$MAE = \frac{\sum_{y_i \in \mathcal{D}_{test}} |y_i - \hat{y}_i|}{|\mathcal{D}_{test}|}, \tag{29}$$

where $\mathcal{D}_{test}$ is the testing set, y_i and $\hat{y}_i$ indicate the ground truth value and the estimated value respectively. The smaller the metric, the better the corresponding performance.

The choice of α has an impact on the reconstruction result. The basic idea of the proposed method is to assign small weighting factors to the frequency contents with higher PSD and large weighting factors to those with lower PSD. However, large or small of the value of the PSD mentioned here is relative. For example, the PSD of the signal s_1 is x_1, and the PSD of the signal s_2 is exactly twice that of x_1, that is, $x_2 = 2x_1$. The weighting factors of the two signals will be $w(x_1) = e^{-\alpha_1 * x_1}$ and $w(x_2) = e^{-\alpha_2 * x_2}$ respectively. In

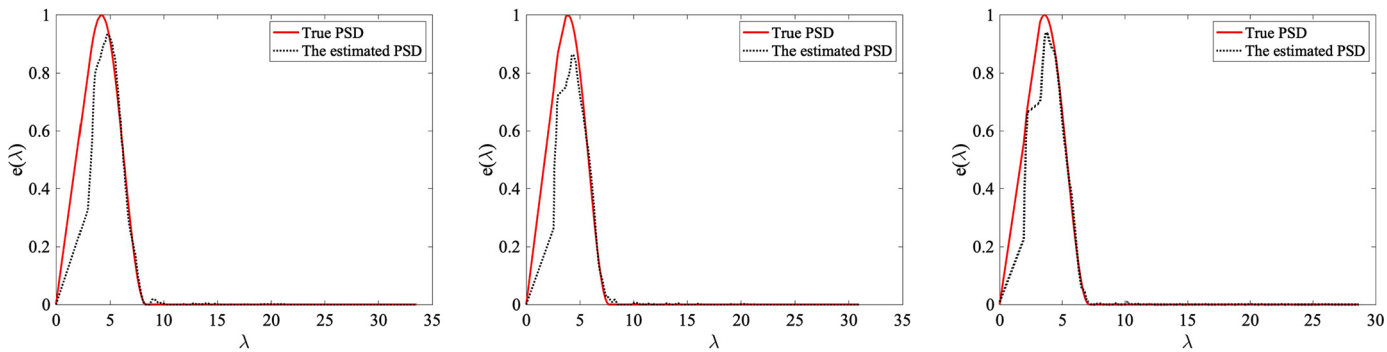

Fig. 4. PSD estimation on graphs with different sizes, from left to right, the numbers of vertices are 1000, 2000 and 5000 respectively.

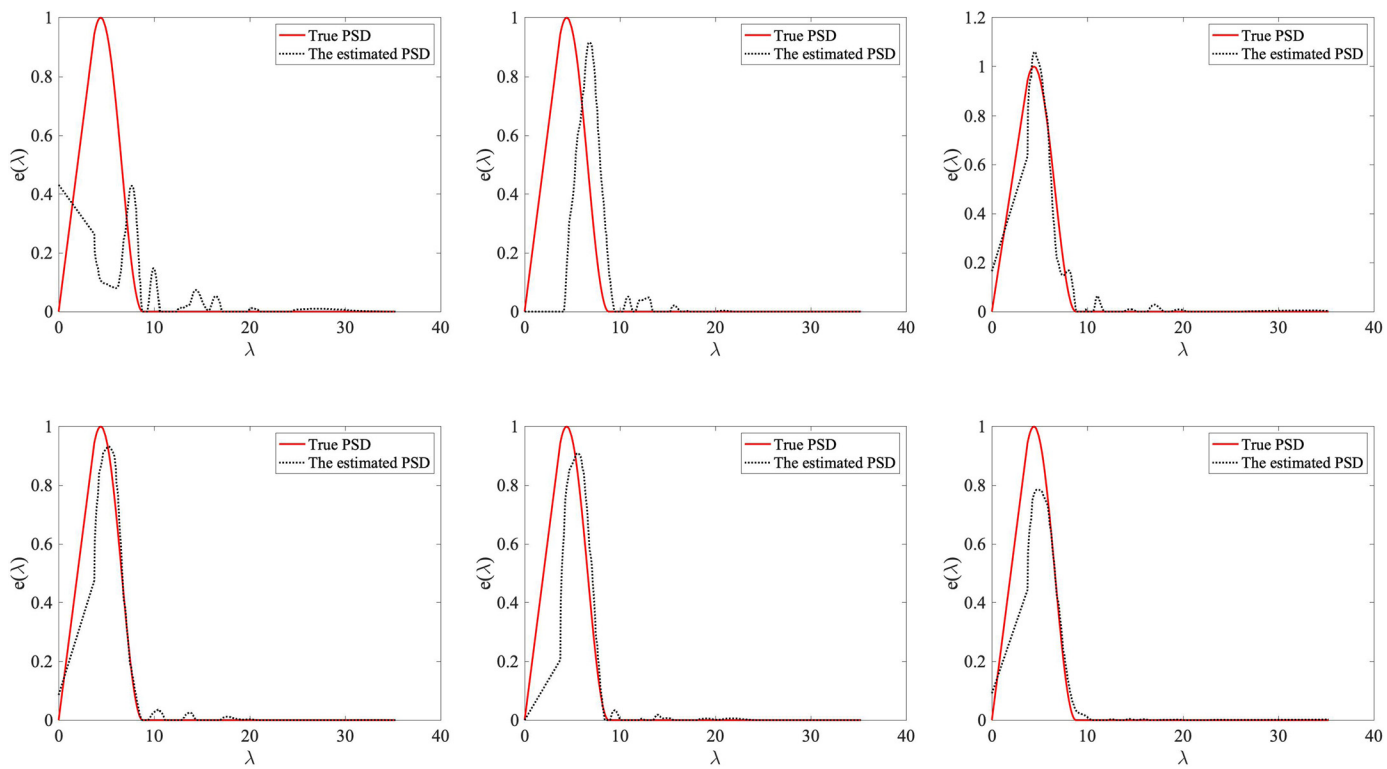

Fig. 5. The influence of the number of sampled signals on PSD estimation. From left to right, from top to bottom, the numbers of sampled signals are 5, 10, 20, 50, 100 and 1000 respectively.

order to get the same weighting factor, that is, $w(x_1) = w(x_2)$, then one must choose α_2 to be half of α_1, that is, $\alpha_2 = \alpha_1/2$. To reduce the influence of the difference in the magnitude of the PSDs of different signals, we adopt the following strategy:

Strategy 1 The maximum value of the PSD is standardized to 1.

In addition to α, another parameter that affects the reconstruction result is the regular parameter β in Equation (14). The first term on the right hand side of Equation (14) is related to the number of labeled vertices, and the second term is related to the connection structure of the graph. This means that the ratio of the labeled vertices, the size of the graph, and the connection structure (such as the number of neighbors) will affect the selection of the parameter β. In order to make the selected β have good universality, we adopt the following strategies:

Strategy 2 Replace the total square error with the average square error of the labeled vertices, and replace the total smoothness with the average smoothness, namely Equation (14) is modified as

$$\min_{\mathbf{f}} r(\mathbf{f}) := \|\mathbf{E}_L\mathbf{f} - \mathbf{y}_L\|^2/|\mathcal{K}| + \beta\mathbf{f}^T\mathbf{U}\Lambda_w\mathbf{U}^T\mathbf{f}/n_w, \tag{30}$$

where $|\mathcal{K}|$ is the number of labeled vertices, and n_w is the number of connections of graph.

Recalling Equation (12), GM directly uses the graph frequency as the weighting factor. If the weighting factor of the proposed method is much larger (or smaller) than the graph frequency, for the same β, the effect of the penalty item is actually different. To facilitate comparison with GM, we employed the following strategy:

Strategy 3 The maximum value of the weight function is standardized to λ_{max}.

The above strategies simplify the choice of parameters. Now, we try to find suitable parameters. We conduct experiments on

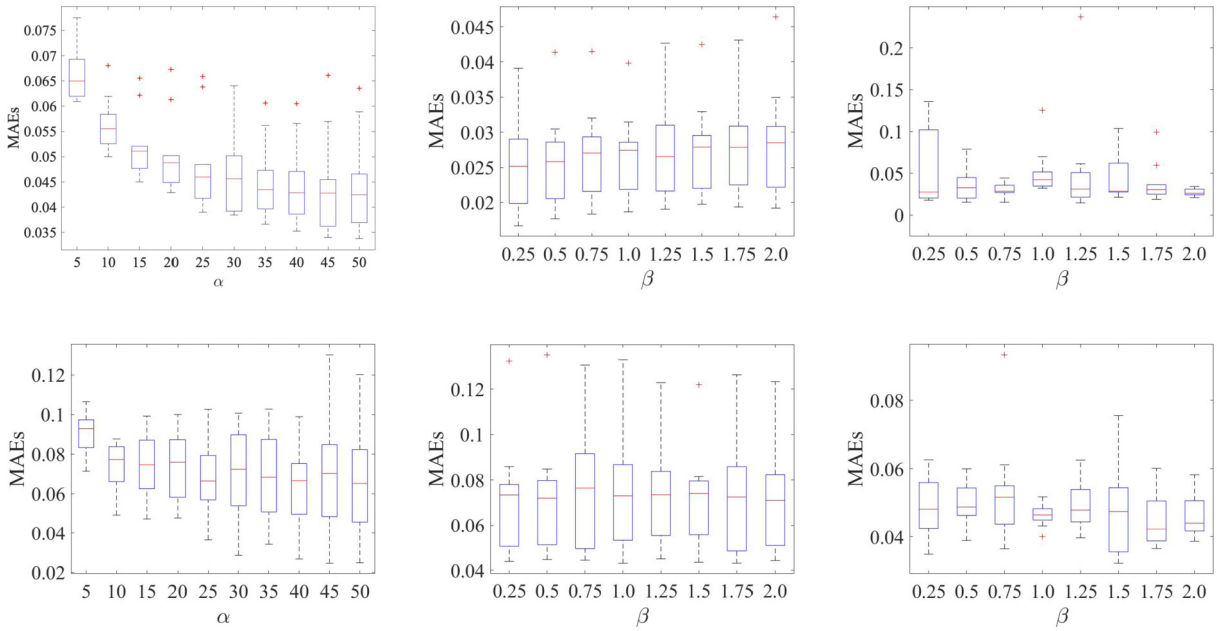

Fig. 6. Top left: PSD is $\gamma_{\mathbf{f}}^1(\lambda_\ell)$. Bottom left: PSD is $\gamma_{\mathbf{f}}^2(\lambda_\ell)$. Top middle: The ratio of labeled vertices is 50%. Bottom middle: The ratio of labeled vertices is 10%. Top right: The number of vertices is 200. Bottom right: The number of vertices is 2000.

Nearest neighbor graphs with 1000 vertices and the number of neighbors 40. Randomly generate 200 signals using the PSD functions of Equation (26) and Equation (27) respectively, and select 10% vertices as the labeled vertices in each signal and the rest of vertices are considered as the unlabeled vertices. All the unlabeled vertices constitute the testing set, and their labels are first saved to be used as the ground truth value when calculating the MAE and then are set to 0 to generate the sampled signals. Take $\beta = 1$, increase α from 5 to 50 with the step of 5, use the proposed model to estimate the labels of the unlabeled vertices of the sampled signals, and calculate the MAE on the testing set. Considering the randomness of the experiment, we repeat the experiment for 10 times. Experimental results are shown at the left panels of Fig. 6. Top corresponds to the PSD function $\gamma_{\mathbf{f}}^1(\lambda_\ell)$, and bottom $\gamma_{\mathbf{f}}^2(\lambda_\ell)$.

Next, we examine the effect of the ratio of labeled vertices on the choice of parameter β. Use the PSD function of Equation (26) to generate 200 random signals on a Nearest neighbor graph with 1000 vertices. Set the ratios of the labeled vertices to 10% and 50% respectively to generate sampled signals and test sets. Take $\alpha = 10$, let β go from 0.25 to 2 in the step of 0.25, repeat the experiment 10 times. Experimental results are shown in the middle panels of Fig. 6.

Finally, we consider the effect of the sizes of graphs on the choice of parameter β. Consider two graphs with different sizes, namely the numbers of vertices are 200 and 2000 respectively. Set the label vertex ratio to 10% to generate sampled signals and test sets. Take $\alpha = 10$, let β change from 0.25 by step size 0.25 to 2, the experiment is repeated 10 times. Experimental results are shown at the right panels of Fig. 6.

We see that:

- When the PSD function is $\gamma_{\mathbf{f}}^1(\lambda_\ell)$, relatively large α bring smaller reconstruction errors, and when the PSD function is $\gamma_{\mathbf{f}}^2(\lambda_\ell)$, except that $\alpha = 5$ gives rise to a slightly worse result, other α do not cause significant difference to the reconstruction results;
- For the different labeled vertex ratios and the distinct graph sizes, within the range considered in the experiments, the values of β do not significantly affect the reconstruction results.

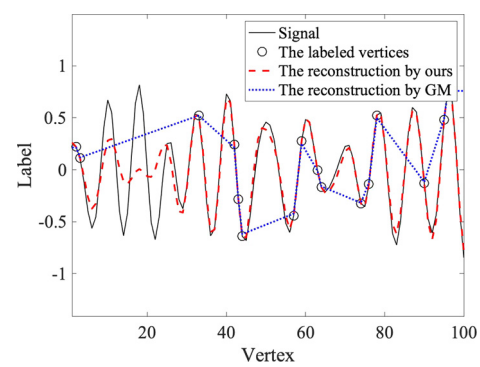

Fig. 7. A reconstruction example on Path graph.

According to the experimental results, in the following experiments, we take the value of α between $10 \sim 50$, and the value of β around 1.

Experiment 5: In order to visually display the reconstruction results, in this experiment, we perform reconstruction experiments on several graphs with small size. The experiments are conducted on the commonly used Path graph, Sensor graph, Nearest neighbor graph (the number of neighbors is 8) respectively. The numbers of vertices are fixed to 100. Use the PSD function of Equation (25) to generate 100 random signals on each graph. Set the label vertex ratio to 20% to generate sampled signals and test sets. The sampled signals are first used to estimate the PSDs, and then the propose method and GM are employed to reconstruct the signals by taking $\alpha = 10, \beta = 0.75$. Fig. 7 and Fig. 8 show several examples. In Fig. 7, the black solid curve is the original signal, the symbols '∘' represent the labeled vertices, the red dotted curve is the signal reconstructed by using the proposed method, and the blue dotted one is the result of GM. We are pleased to see that the reconstructed results by using the proposed method are obviously better than those by using GM. When using the proposed method, the reconstructed signal values at most of vertices are very close to those of the original signal, and intuitively, the reconstructed signal's waveform fit that of the original signal very well, while GM

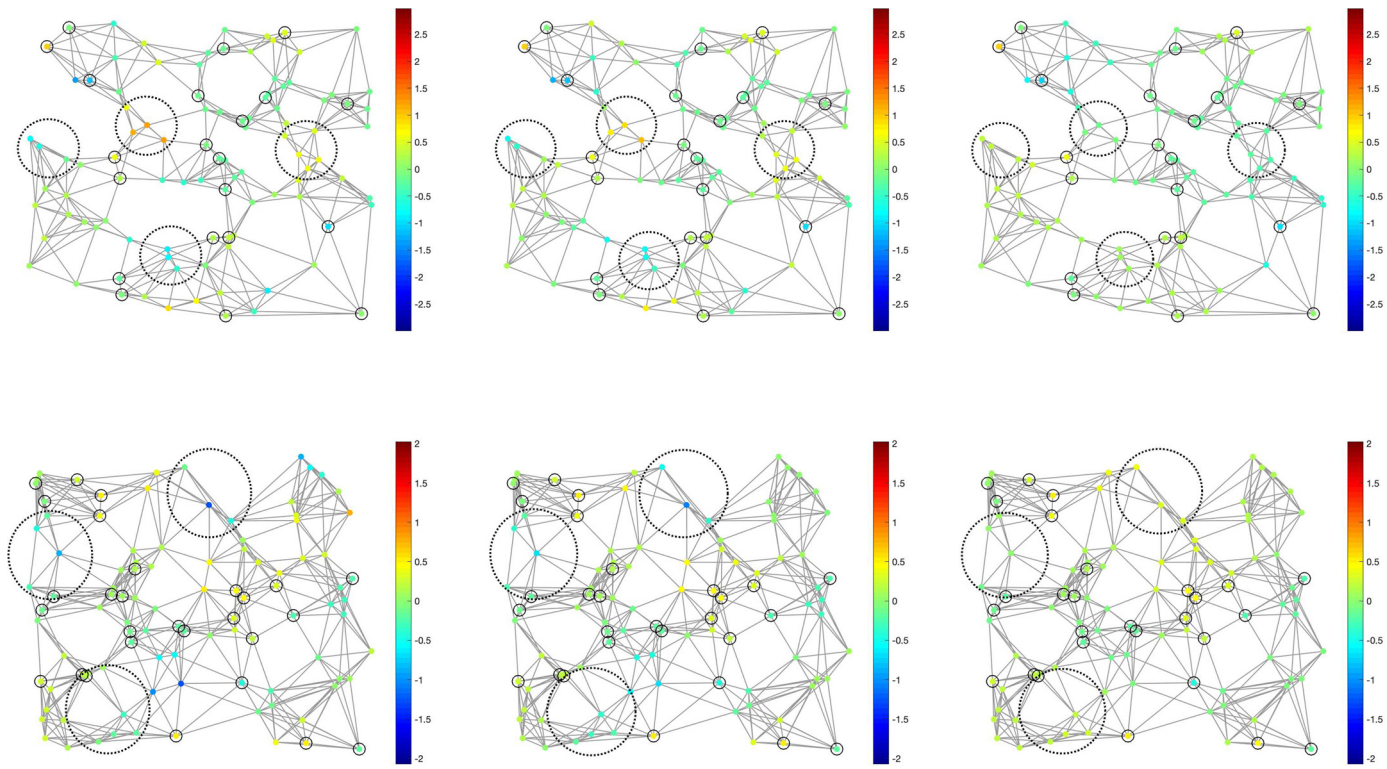

Fig. 8. The reconstruction examples on Sensor graph and Nearest neighbor graph. Top: Sensor graph. Bottom: Nearest neighbor graph. Left: The original signals. Middle: The reconstructed signals by using the proposed method. Right: The reconstructed signals by using GM.

Table 1
MAEs on three graphs.

	Path	Sensor	NN
GM	0.3195	0.2703	0.2295
The proposed method	0.1489	0.1541	0.1039

fails to do these. Fig. 8 shows the examples on Sensor graph and Nearest neighbor graph. The symbols '∘' represent the labeled vertices. The top corresponds to Sensor graph, and the bottom Nearest neighbor graph. The left, middle and right panels correspond to the original signal, the reconstructed signals by using the proposed method and those by using GM. Some key areas are marked with dotted circles to facilitate comparison. The signals reconstructed by using our method are very close to the original signals, while the results by using GM are much worse. We also calculated the MAE on testing set, and the results are shown in Table 1. The results on all the three graphs show that the reconstruction performance of the proposed method is significantly better than that of GM.

Experiment 6: The experiments are performed on Nearest neighbor graphs, and the proposed method and GM are comprehensively compared by changing the ratio of labeled vertices, the number of neighbors, the number of sampled signals, and the graph size. Experiments are divided into four groups. The parameters of each group are shown in Table 2. All experiments use the PSD function of Equation (25) to generate the original random signals. Extract randomly vertices as the labeled vertices in given ratios, and the rest vertices are regarded as the unlabeled vertices. All the unlabeled vertices constitute the testing set, and their labels are first saved to be used as the ground truth value when calculating the MAE and then are set to 0 to generate the sampled signals. Each sampled signal set is first used to estimate the PSD, and then the propose method and GM are employed to reconstruct signals by taking $\alpha = 10, \beta = 0.75$. Taking into account the randomness of the experiments, each group is performed independently for 10 times. Experimental results are shown in Fig. 9, where the left panels are the results of the proposed method, and the right panels are the results of GM, from top to bottom correspond to $Group1 \sim Group4$. It can be summarized as follows:

1. The MAE decreases as the proportion of labeled vertices increases, but in all cases, the reconstruction errors of the proposed method are significantly lower than that of GM. The average MAE of the proposed method when the labeled vertex ratio is 10% is even smaller than that of GM when the labeled vertex ratio is 75%.
2. Both methods show that the smaller the number of neighbors, the worse the reconstruction performance. Especially when the number of neighbors is 10, the result of the proposed method is not even as good as that of GM. But after the number of neighbors is greater than 20, the average MAE of the proposed method is significantly lower than that of GM.
3. GM does not need to estimate PSD, so the number of sampled signals does not affect the reconstruction errors, which can be seen from the result on the right panel of the third row of Fig. 9. When the number of sampled signals is 1 and 2, the PSD can hardly be estimated accurately, so there are large estimation errors. But after the number of sampled signals increases to 10, the reconstruction performance tends to be stable, and the average MAEs are significantly smaller than those of GM.
4. Regardless of the size of graphs, the reconstruction performance of the proposed method is significantly better than that of GM. There does not seem to be a causal relationship between the size of graphs and the reconstruction performance. Although the experimental results show that the MAE is the largest when the number of vertices is 500, we believe that this result is due to the randomness of sampled signals and connection structures of the graph.

Table 2
Parameter configuration of experiment 7.

	Fixed			Variable
	Vertex	Signal	Neighbor	Percentage (%)
Group 1	500	1000	40	75, 50, 20, 10, 5
	Vertex	Signal	Percentage (%)	Neighbor
Group 2	500	1000	10	10, 20, 30, 40, 50, 60
	Vertex	Neighbor	Percentage (%)	Signal
Group 3	500	40	10	1, 2, 5, 10, 50, 100, 1000
	Signal	Neighbor	Percentage (%)	Vertex
Group 4	1000	40	10	100, 200, 500, 1000, 2000, 5000

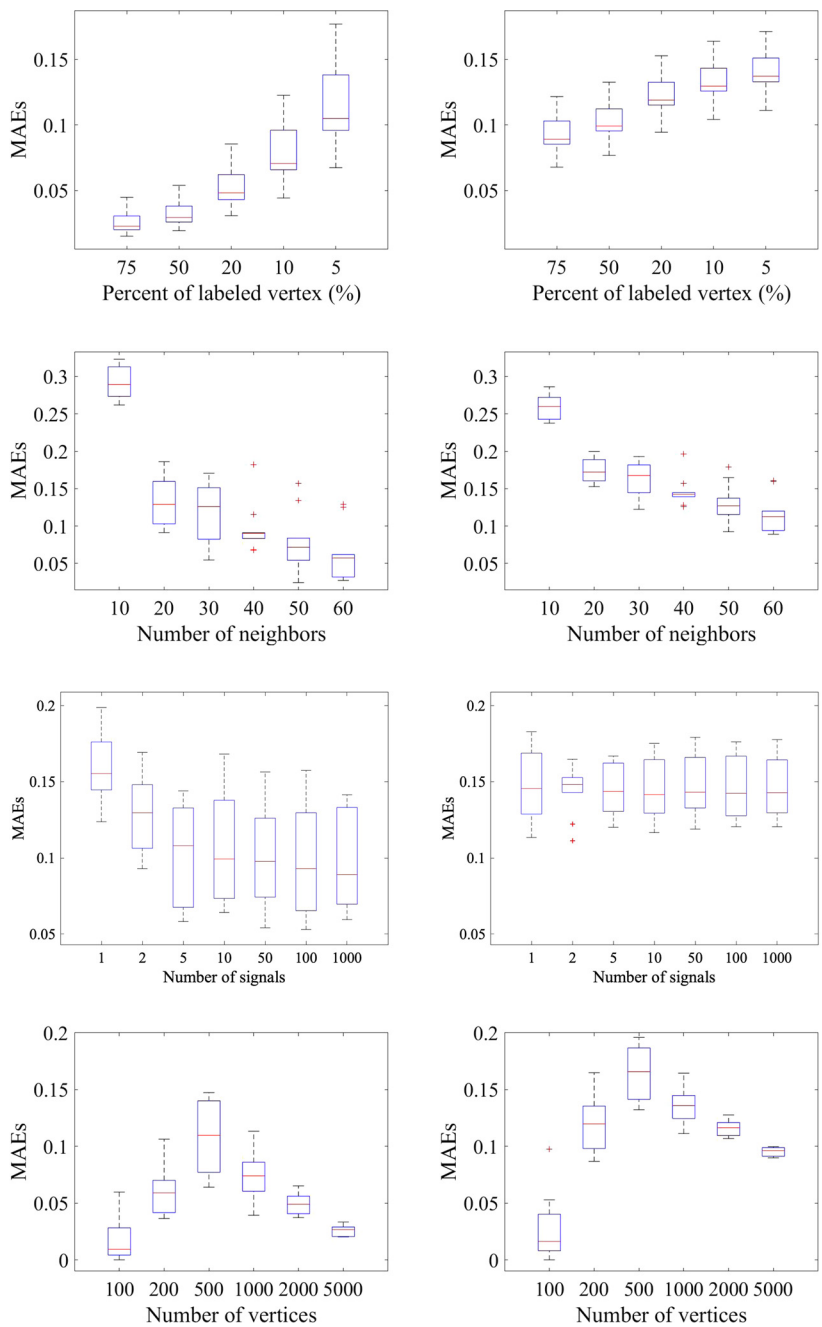

Fig. 9. Results of Experiment 7. Left: Our method. Right: GM. From top to bottom correspond to *Group*1 ∼ *Group*4.

Experiment 7: This experiment is carried out on the public face data set ORL. This data set includes 400 face grayscale images with the size of 112×92. In the experiment, we randomly selected the pixels of the image in the ratio of 50%, 60%, 70%, 80%, 90% as the unlabeled vertices and set their values to 0, and then cut each image into disjoint small blocks of 8×8. Since the image width is not an integral multiple of 8, we remove the two pixels on the left and right in advance, so that each image can be cut into 154

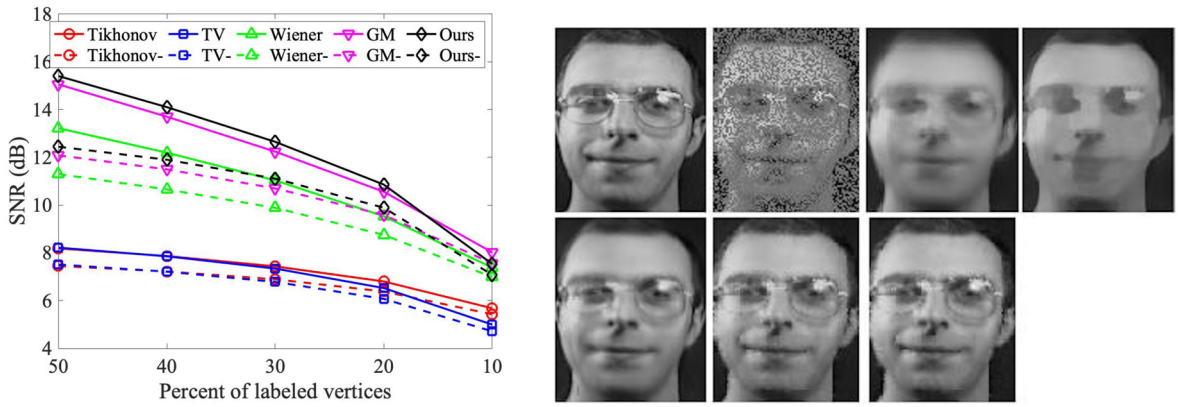

Fig. 10. Reconstruction experiment on ORL face dataset. Left: The SNRs of the reconstruction results of the five methods. The solid lines represent including all vertices, and the dotted lines represent including only the unlabeled vertices. Right: An example when randomly extracting 50% pixels as the labeled vertices. In order, from left to right and from top to bottom correspond to the original image, the noisy image and the reconstructed images by Tikhonov (SNR: 11.1773 dB), TV (SNR: 11.2354 dB), Wiener filtering (SNR: 16.1859 dB), GM (SNR: 16.2281 dB) and ours (SNR: 16.8504 dB).

image blocks. For each small block of 8×8, we use the position coordinates of each pixel as the feature vector to create a Nearest neighbor graph with 64 vertices and the number of neighbors 8. For each image, use its graph sampled signals corresponding to 154 image blocks to estimate the PSD and reconstruct the sampled signals with the parameters $\alpha = 20, \beta = 0.75$. Taking the original image as the reference, two SNRs (that is, the labeled vertices are included or not), and then the average SNR of all images is calculated. The result is compared with those by using GM and other three methods (Tikhonov, TV and Wiener filtering) reported by [34]. Experimental results are shown in Fig. 10, where the left panel is the SNRs of five methods, the abscissa is ratio of the labeled vertices, the ordinate is SNR, the solid line indicates the case of including all vertices and the dashed line represents that of including only the unlabeled vertices. The right panel shows an example. The two images on the left of the first row are the original image and the noise image. Others are the reconstructed images by five methods. We see that among the five labeled vertex ratios considered in the experiment, the average SNR of the proposed method is usually better than those of other methods, although this advantage gradually decreases as the labeled vertex ratio decreases.

In particular, it should be noted that the experiments of the reference methods used 300 images to construct the Nearest neighbor graph, and then 100 ones of which are used to estimate the PSD. These images are all original images (all the vertices are the labeled vertices). Finally the remaining 100 ones of 400 images are used to test. Our method only uses the image to be reconstructed itself and does not need any other original images to estimate the PSD. Therefore, the proposed method is particularly meaningful for the applications where multiple original images of similar scenes cannot be obtained.

We compare experimentally the computational times of five methods. The experiments are conducted on MacBook Pro with 2.6 GH Intel 2-Core i5 CPU, 8 GB 1600 MHz DDR3 memory. MATLAB version is 2018a. In Tikhonov and Wiener, we use 300 images to generate a Nearest neighbor graph by employing GSPbox [48], which costs 40.27s. Next, Wiener needs to spend 345.95s to estimate the PSD. Because these operations only need to be performed once, the above time is not included in the statistics below. We then calculate the average of the running time of each algorithm on 100 images. The results corresponding to Tikhonov, TV and Wiener are 0.8946s, 7.0651s and 6.7437s, respectively. We also calculated the average running times on 100 images using GM and the proposed method. It shows that they are only 0.0334s and 0.0361s for GM and the proposed method, respectively. Moreover, these have contained all the times for establishing the nearest neighbor graphs, estimating the PSDs (only for the proposed method) and reconstructing signals.

Experiment 8: The recommendation system (RS) has been widely used in major e-commerce websites. The key technology of recommendation systems is to predict unknown ratings based on the ratings of a small number of users on items. Let $\mathcal{U}$ and $\mathcal{I}$ be the sets of the users and the items respectively in a RS, then the ratings of all the users to all the items form a matrix $\mathbf{R} \in \mathcal{R}^{|\mathcal{U}| \times |\mathcal{I}|}$, where $|\cdot|$ indicates the cardinality of a set, and the element $r(i, j)$ represents the rating by the i-th user to the j-th item. In practice, a user often does not rate all items and an item is also not rated by all users. Hence the matrix $\mathbf{R}$ is usually sparse. The goal of prediction techniques is to predict the unknown ratings in the rating matrix $\mathbf{R}$. This is essentially a signal reconstruction problem. Therefore, the last experiment of this paper is designed to predict unknown ratings in recommendation system.

When modeling the prediction as a graph signal reconstruction problem, we can either use items or users as the vertices of the graph. When users serve as vertices, all users' ratings for an item are regarded as a graph signal. Similarly, when items serve as vertices, a user's ratings for all items is a graph signal. For convenience, we will refer to them as user-based reconstruction and item-based reconstruction in the following text.

The famous MovieLens-100k data set [49] is employed to experiment. Readers can access http://files.grouplens.org/datasets/movielens/ for details. The data set consists of 100,000 ratings ($1 \sim 5$) from 943 users on 1682 movies. Each user has rated at least 20 movies. The experiment is carried out on 5 pairs of training and testing sets, namely $u1.base \sim u5.base$ and $u1.tests \sim u5.test$. Each pair of *base* and *test* is a 80%/20% splits of the whole dataset. There are disjoint test sets between any two pairs. In the training sets, the proportion of the known labels is about 5%.

We follow the method in the literature [50] to extract the feature vector of each user (user-based) and item (item-based) from the training data to build Nearest neighbor graphs. In the user-based experiment, The parameters are set as follows: The number of neighbors is 30, $\alpha = 20, \beta = 0.75$ for user-based prediction and the number of neighbors is 50, $\alpha = 35, \beta = 0.25$ for item-based situation. Use the proposed method to estimate the labels of the unlabeled vertices of the training data, compare them with the true values in the test set, calculate the MAE, and finally calculate the average of the MAEs on the five sets of data. The results are shown in Table 3, in which the results of several reference methods are also given. Although our results are slightly inferior to

Table 3
The comparison with the popular methods.

Methods	UB_{BCF} [54]	UB_{HUS} [55]	$SVD++$ [56]	PMF [52]	BPMF [53]
Ave. MAE	0.7751	0.7870	0.7864	0.7370	0.7614
Std.	0.0042	0.0019	0.0069	0.0082	0.0047
Methods	FPMF [51]	RHKS+kBR [50]		Ours	
		user-based	item-based	user-based	item-based
Ave. MAE	0.7318	0.7546	0.7314	0.7568	0.7403
Std.	0.0088	0.0056	0.0054	0.0060	0.0051

the latest two results [50–52], they are better than other methods, indicating that the proposed method is effective for the recommendation problem. In particular, the methods of the literatures [50–53] belong to the fusion technology of multiple methods. If compared with the results of the single technologies [54–56], our results are much better.

4. Conclusion

This paper discusses the problem of signal reconstruction on graphs. A reconstruction model based on power spectral density estimation of stationary stochastic graph signals is proposed, and at the same time an algorithm to estimate the power spectral density directly from sampled signals is given. Some well-designed experiments verify the effectiveness of the power spectral density estimation algorithm. Experiments on the simulation data sets prove that the reconstruction performance of the proposed reconstruction model is significantly better than that of the general model. Experimental results on two real data sets show that the results of the proposed model are usually better than those of state-of-the-art methods.

CRediT authorship contribution statement

Zhihua Yang: Conceptualization, Methodology, Writing – original draft. **Guangrui Yang:** Data curation, Software. **Lihua Yang:** Funding acquisition, Writing – review & editing. **Qian Zhang:** Funding acquisition, Project administration, Software, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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